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IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-439

1. Topic (12 words max)

An AI-Powered Cognitive Screening & Analysis Platform

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Dementia is a progressive neurological disorder marked by the decline of memory, reasoning, and behavioral functioning, posing significant challenges to daily living and long-term independence. According to the World Health Organization (2021), over 55 million individuals globally are currently affected, with nearly 10 million new cases reported annually. In Sri Lanka, the rising elderly population, combined with a lack of affordable and accessible clinical screening tools, underscores the critical need for culturally appropriate, early diagnostic technologies. Traditional clinical assessments such as MRI scans or MMSE-based evaluations are often inaccessible due to financial, infrastructural, and awareness barriers—particularly in rural areas.

To address these challenges, this research proposes a comprehensive, mobile- and web-based dementia screening platform tailored to the Sri Lankan population. The system integrates multiple non-invasive, low-cost approaches powered by machine learning and AI. A key innovation is the use of Sinhala handwriting stroke features—collected via touchscreens or stylus input—which are analyzed for cognitive markers such as pen speed, pressure, jerk, and stroke frequency. These digital biomarkers have shown strong correlation with cognitive impairments in previous research (Giancardo et al., 2016; Drotár et al., 2014). The system also incorporates a custom MMSE-style screener and structured lifestyle questionnaire to gather data on sleep, diet, memory complaints, and mood.

Unlike conventional tools, this project emphasizes local language support (Sinhala), minimal hardware requirements, and real-time usability—features critical for adoption in low-resource settings. It also aims to improve clinical triage accuracy and raise awareness by allowing at-risk individuals to self-screen from home. By combining AI with culturally sensitive design, this tool can fill a vital gap in dementia prevention, especially in developing regions.

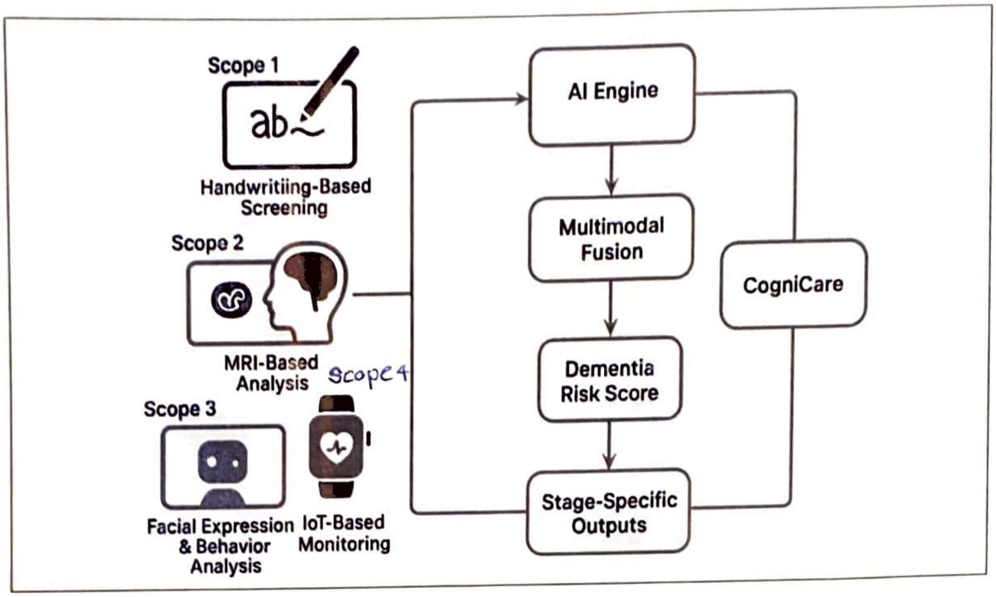
Key References

- World Health Organization. (2021). *Dementia Fact Sheet*. https://www.who.int/news-room/fact-sheets/detail/dementia
- Giancardo, L., Sánchez-Ferro, Á., Arroyo-Gallego, T., Butterworth, I., Mendoza, C. S., & Obeso, J. A. (2016). *Computer keyboard interaction as an indicator of early Parkinson's disease*. Scientific Reports, 6, 34468.
- Drotár, P., Mekyska, J., Rektorová, I., Masarová, L., Smékal, Z., & Faundez-Zanuy, M. (2014). *Evaluation of handwriting kinematics and pressure for differential diagnosis of Parkinson's disease*. Artificial Intelligence in Medicine, 67, 39–46.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is **CogniCare**, a comprehensive, AI-powered dementia screening platform designed for web and mobile use. It addresses the growing challenge of late or inaccessible dementia diagnosis in Sri Lanka, especially in low-resource and rural communities. The system leverages multiple **non-invasive data sources**-including digital handwriting input, MRI scans, facial and behavioral cues, and wearable device data-to detect early cognitive decline. Users interact with a Sinhala-localized interface where they can perform simple tasks such as drawing characters, answering lifestyle questions, or completing in-app activities.

Simultaneously, background systems analyze data like pen pressure, stroke speed, facial expressions, idle behavior, and physiological patterns. Clinical data such as MRI brain scans can be uploaded for deeper triage when available. All collected inputs are processed by a centralized **AI engine**, which performs **multimodal fusion** to produce a dementia risk score and stage-specific output. The system offers real-time feedback and can guide the user or caregiver toward appropriate next steps, such as seeking clinical advice or following cognitive wellness recommendations. Designed with privacy, scalability, and cultural sensitivity in mind, CogniCare aims to democratize access to dementia screening, reduce the burden on healthcare professionals, and promote early detection ultimately contributing to better long-term outcomes for aging populations.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Domain Expertise Required:

- Cognitive neuroscience and behavioral psychology: To understand dementia progression, including indicators like memory loss, disorientation, and motor disturbances.
- Clinical neurology and radiology: For interpreting handwriting stroke patterns and MRI brain scans (e.g., hippocampal shrinkage, cortical thinning).
- Health informatics and geriatric care: To translate diagnostic criteria (MMSE, MoCA) into automated digital screening mechanisms.
- User experience for vulnerable groups: To design safe, accessible, and culturally sensitive interfaces for elderly users.

Technical Knowledge Required:

- Computer Vision & Deep Learning: Used for analyzing T1-weighted MRI scans to detect structural changes in the brain associated with dementia. Techniques include convolutional neural networks (CNNs) such as ResNet and VGG, trained to classify dementia stages based on features like hippocampal atrophy and cortical thinning.
- Signal Processing & Stylometry: Essential for processing handwriting input from Sinhala users via Wacom tablets or touchscreen devices. Extracted features include pen pressure, stroke velocity, pen lift counts, and jerk, which are linked to fine motor skills and cognitive function.
- AI/ML Modeling: Machine learning algorithms such as Decision Trees, Logistic Regression, Isolation Forests, and Support Vector Machines are used for classifying dementia risk, detecting behavioral anomalies, and identifying progression stages. These models operate on multimodal input including smartwatch data, cognitive test results, and user engagement trends.
- Emotion & Facial Analysis: Real-time emotion detection during cognitive tasks is achieved using tools like MediaPipe, OpenCV, and Affectiva, enabling the identification of stress, confusion, or apathy—key markers in dementia risk scoring.
- Wearable API Integration: Technologies such as the Fitbit Web API is employed to collect continuous physiological signals including heart rate variability, sleep data, and step count.
- Web Application Development: Interactive and accessible UIs are developed using React for the frontend and Python (Flask) or Node.js for backend services. These interfaces support screening tests, monitoring dashboards, doctor-patient interaction modules, and personalized recommendation delivery.

Data Requirements:

- Handwriting data: Sinhala stroke samples with cognitive labels (MMSE-based).
- MRI scans: Public datasets like OASIS and ADNI, labeled with dementia stages.
- Physiological data: Heart rate, step count, and sleep logs from smartwatches.
- Facial expression/emotion metrics: Micro-expressions during tasks.
- Behavioral logs: Task completions, missed alerts, idle time, and help requests.
- Cognitive performance: Memory task outcomes, reaction times, missed reminders, repeated attempts
- Clinical datasets: TIHM, BESI, and CAPTURE-24 for training and benchmarking risk models
- User engagement data: Help clicks, idle periods, notification response delay
- Doctor Feedback & Notes: helps ,final risk validation and clinical decision-making.

1. Objectives and Novelty

Main Objective			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Senaratne. H.D It22358998	To develop a lightweight and accessible digital screening tool for early detection of dementia in Sri Lanka by integrating handwriting analysis, MMSE-style cognitive scoring, and lifestyle factors into a unified machine learning model.	1. Collect and preprocess data from handwriting stroke datasets and MMSE/lifestyle inputs. 2. Design a handwriting input interface (mobile, Wacom, or touchscreen compatible). 3. Implement a custom MMSE-style screener to extract cognitive scores. 4. Develop a lifestyle questionnaire module to capture sleep, diet, memory complaints, and related factors. 5. Extract stroke features such as pressure, pen-down events, mean speed, jerk, and gesture ratios. 6. Fuse all features (handwriting, MMSE scores, lifestyle inputs) into a structured dataset.	**Sinhala handwriting analysis module** that captures stroke-based features like speed, pressure, jerk, and pen-down events using touchscreen or Wacom input. **Custom MMSE-style cognitive screener**, culturally adapted and freely usable, assessing domains such as memory, attention, and orientation. **Lifestyle and cognitive health questionnaire** covering memory complaints, sleep quality, diet, mood, and

		7. Train baseline machine learning models like Random Forest and XGBoost for initial prediction. 8. Compare model performance by testing different feature subsets and model architectures. 9. Prototype a simple user interface (UI) for user input and displaying dementia risk output. 10. Analyze model interpretability and highlight key factors contributing to dementia predictions.	confusion to enhance screening accuracy. **Mobile-compatible dementia screening tool** designed for the Sri Lankan population, accessible via smartphones or tablets. **Early fusion AI model** that combines handwriting features, MMSE scores, and lifestyle inputs to predict dementia likelihood (Yes/No). **Machine learning pipeline** that includes Random Forest, XGBoost, and DNN, with model comparison and performance evaluation. **Novelty lies in** fusion of handwriting + lifestyle + cognitive inputs, and low-cost, non-invasive deployment — not previously done in Sri Lanka.
Dissanayake M.P.S. IT22901644	To detect and classify the stage of dementia using MRI brain scans, enabling	1. Preprocess MRI images (resizing, normalization, augmentation). 2. Train a CNN or transfer learning model (e.g., VGG16, ResNet) using labeled MRI datasets.	Moves beyond binary dementia detection to **multi-class classification**, providing stage-specific output.

	automated, accurate, and non-invasive diagnosis across four categories: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented.	3. Input MRI image and predict one of four dementia stages. 4. Evaluate performance using metrics such as accuracy, precision, recall, and confusion matrix. 5. Optionally visualize brain regions influencing predictions using Grad-CAM or saliency maps.	Uses AI for **fully automated MRI analysis**, reducing dependency on radiologists. Supports **early-stage intervention** by identifying Very Mild Dementia. Enables repeatable and scalable screening in clinical or research settings using open datasets.
Natheer H A H IT22888952	To develop a facial expression and behavioral pattern-based dementia risk detection module by leveraging in-app usage behavior and real-time webcam-based emotional analysis to passively detect cognitive anomalies and flag early signs of dementia.	• Integrate facial emotion recognition using webcam-based APIs (MediaPipe, face-api.js) • Track task behavior logs: idle time, missed notifications, help clicks, response delays • Log behavioral events silently during memory or attention tasks • Analyze confusion, apathy, or emotional flatness using emotion sequences • Compute behavioral indicators like "Task Disruption Index" and "Cognitive Function Deviation Score" • Combine emotion data + behavior logs into a feature set • Train ML models (Logistic Regression / Random Forest) to predict risk • Design real-time alert system when behavior crosses thresholds • Build frontend popups to suggest screening or notify caregivers • Ensure privacy-compliant, consent-based emotion capture with ethical guidelines	• First Sri Lankan research system combining facial expressions + behavior tracking for early dementia detection via basic webcams and app logs • Non-invasive, passive, and real-time emotional-behavioral monitoring during digital tasks • Creation of a dynamic cognitive risk scoring model using usage data + emotion events • Bridges psychology, UI/UX, and AI — enabling emotion-aware health tech • No need for clinical devices — only smartphone/laptop webcam and app interaction • Culturally aligned interface with ethical consent handling for emotional sensing

Piruthivi K IT22358820	Collect physiological data Connect to smartwatch API, Monitor daily behavior, Compare with clinical patterns Detect anomalies, Generate wellness suggestions, Support cognitive engagement Sync with dementia staging, Mental well-being ,activities	• Integrate Fitbit Collect Heart rate, sleep, Blood Pressure . • Track routines, inactivity, sleep, and movement trends Use ML models to detect unusual patterns in app usage and health data • Deliver alerts and health suggestions based on behavior • Feed behavioral trends to stage prediction engine • Recommend engaging activities: music therapy, puzzles, meditation, coloring tasks	• This is a **non-invasive, AI-based early screening tool,** logs for early-stage cognitive screening, • It does not depend on questionnaires or games to detect problems . it uses **real physiological patterns**, making it more passive and accessible. real-time detection of cognitive anomalies using non-intrusive sensors (e.g., heart rate, sleep, motion), • provide personalized, daily recommendations to encourage healthy routines and mental stimulation which is essential for dementia prevention.

2. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Senaratne. H.D It22358998	This scope aims to build a mobile- and web-compatible dementia screening tool tailored for the Sri Lankan context. It fuses handwriting stroke features, cognitive test scores, and lifestyle data to predict early signs of dementia using machine learning models. The tool collects Sinhala handwriting via touchscreen or Wacom, calculates MMSE-style scores through a custom screener, and gathers lifestyle and behavioral risk factors through questionnaires. All inputs are fused using early fusion techniques to train classifiers like Random Forest and XGBoost, enabling a lightweight, explainable, and culturally localized diagnostic aid.
Dissanayake M.P.S. IT22901644	This scope focuses on the **automated detection and classification of dementia stages using MRI brain scans**. By leveraging deep learning techniques typically convolutional neural networks the system analyzes structural brain patterns to determine whether a patient is **Non-Demented, Very Mild Demented, Mild Demented**, or **Moderate Demented**. The MRI image is uploaded into the model, which then processes it to identify early signs of brain atrophy or other dementia-related changes, outputting the exact dementia level. This approach supports early, non-invasive diagnosis and helps clinicians make more informed decisions by providing stage-specific results. It can be a powerful tool for supplementing traditional clinical assessments with precise imaging-based evidence.
Natheer H A H IT22888952	My component focuses on developing a facial expression and behavioral pattern-based dementia screening module using a webcam and in-app activity tracking. It captures emotions like confusion or apathy during cognitive tasks using tools like MediaPipe or face-api.js, while silently logging user behavior such as idle time, missed tasks, and repeated clicks. These features are used to compute a cognitive risk score through machine learning models like Logistic Regression or Random Forest. This aligns with my specialization in AI-integrated web systems and intelligent human-computer interaction, enabling real-time, privacy-aware, and non-invasive dementia risk monitoring.
Piruthivi K IT22358820	It will be responsible for implementing the IoT-based (using smart watch)data collection and behavioral analysis system. My component will connect with wearable APIs (e.g., Fitbit), collect physiological and behavioral data (heart rate, sleep, step count, inactivity), and use machine learning to detect cognitive anomalies. I will also develop a recommendation engine that suggests cognitive games, music therapy, and daily engagement tasks to improve mental health and early dementia management. This scope aligns with my specialization in AI and IoT-based health systems

3. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mrs.	Manori	Gamage	MGamage
Co-Supervisor	Ms.	Kaushalya	Rajapakse	KR. 2025/06/26
External Supervisor	Dr.	Malsha	Gunathilaka	

Summary of external supervisor's (if any) experience and expertise:

Dr.Malsha, a Consultant Old Age Psychiatrist, presently serves at the Psychiatry Unit at Colombo South Teaching Hospital, Kalubowila, Sri Lanka. With a keen focus on advancing evidence-based interventions for the elderly population with cognitive impairment in Sri Lanka, Malsha took the initiative to launch the Cognitive Stimulation Therapy (CST) project alongside her team. Currently, she oversees CST groups at Kalubowila Hospital and also fulfills the role of a CST trainer within Sri Lanka.

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Topic Assessment Form

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Recomended.

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Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important**:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.